

Jasper Zhang

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EDUCATION

William A. Shine Great Neck South High School

Great Neck, NY

Class of 2027 | GPA: 100.5/100 (weighted)

Sep. 2023 – Jun. 2027

- **Coursework:** AP Calculus BC, AP Chemistry, AP Physics 2, AP US History, AP English Language.
- **Honors:** U.S. Earth Science Olympiad Top 45 National Finalist (2026); U.S. Presidential AI Challenge New York State Champion (2026); ASM International Special Research Award (2025); AIME Qualifier (2025); Long Island Young Mathematics Scholar Award (2024)

EXPERIENCE

Machine Learning & Computational Biology Researcher

2025 – Present

Great Neck South Research Facility

Great Neck, NY

- Lead first/second author across six research papers in machine learning for biology, working on problems spanning mechanistic interpretability, multi-task learning theory, and molecular and genomic property prediction; accepted to ICLR 2026 workshops with 16 total paper acceptances and 2 oral presentations (ICLR GEM Workshop, LION20).

Electrochemistry Researcher

Summer 2025

Tufts University — Leverick Group

Medford, MA

- Led an independent lithium-ion battery project under Prof. Leverick, designing cell chemistries and fabricating full cells from electrode preparation through assembly; applied scanning electron microscopy (SEM) and related characterization techniques to analyze electrode/electrolyte microstructure and post-cycling degradation.

Biochemistry Research Assistant

Summer 2024

Stony Brook University — Tonge Group

Stony Brook, NY

- Contributed to ongoing development of small-molecule inhibitors targeting enzyme drug targets implicated in cancer pathways; cultured E. coli for recombinant protein expression and gene extraction, performing spectrophotometry, plasmid isolation, column chromatography, and culture incubation.

PROJECTS

*denotes co-first authorship.

The Mechanistic Invariance Test: Genomic Language Models Fail To Learn Positional Regulatory Logic.

J. Zhang*, B. Cheng*. *ICLR 2026 GEM Workshop (Oral)*.

- Built a benchmark probing whether genomic language models learn positional regulatory logic; evaluated five major architectures spanning autoregressive, masked, and state-space paradigms, and showed that apparent positional sensitivity is driven by AT-content shortcuts, with a 100-parameter baseline outperforming billion-parameter models.

Single-Position Intervention Fails: Distributed Output Templates Drive In-Context Learning.

J. Zhang*, B. Cheng*. *LION20 (Oral)*.

- Used causal activation interventions to show that in-context task identity is distributed across demonstration tokens rather than localized; single-position edits achieve 0% transfer while multi-position edits reach up to 96% across LLaMA, Qwen, and Gemma.

Information-Theoretic Requirements for Gradient-Based Task Affinity Estimation in Multi-Task Learning.

J. Zhang*, B. Cheng*. *ICLR 2026 GEM Workshop*.

- Identified a sharp phase transition in gradient-based task affinity estimation around ~30–40% sample overlap; showed that benchmarks like MoleculeNet and TDC operate far below this threshold, explaining years of inconsistent multi-task learning results.

TECHNICAL SKILLS

Programming & ML: Python, Java; PyTorch, NumPy, pandas, scikit-learn, Matplotlib, LaTeX; deep learning, GANs, CNNs, language models, multi-task learning, mechanistic interpretability

Lab & Field Techniques: SEM, spectrophotometry, plasmid isolation, column chromatography, cell culture, gene extraction, recombinant protein expression, lithium-ion battery fabrication

Languages: English (native), Mandarin Chinese